light to travel the distance $2d$ is calculated from the angular speed ω of the toothed-wheel.

The angular speed ω (with unit rad s^{-1}) of the toothed-wheel when the light disappeared for the first time is,

$$\omega = \frac{\theta}{t} \qquad (6.13)$$

Here, θ is the angle between one tooth and the next slot which is turned within that time t.

$$\theta = \frac{\text{total angle of the circle in radian}}{\text{number of teeth+number of cuts}}$$

$$\theta = \frac{2\pi}{2N} = \frac{\pi}{N}$$

Substituting θ in equation (6.13),

$$\omega = \frac{\pi / N}{t} = \frac{\pi}{Nt}$$

Rewriting the above equation for t,

$$t = \frac{\pi}{N\omega} \qquad (6.14)$$

Substituting this in equation (6.12),

$$\upsilon = \frac{2d}{\pi / N\omega}$$

After rearranging,

$$\upsilon = \frac{2dN\omega}{\pi} \qquad (6.15)$$

Fizeau had some difficulty to visually estimate the minimum intensity of the light when it is blocked by the adjacent tooth. The value of speed of light determined by him was very close to the actual value. Later on, with the same idea of Fizeau and with much sophisticated instruments, the speed of light in air was determined as, $\upsilon = 2.99792 \times 10^8$ m s^{-1}.

Note

After the disappearance of light for the first time while increasing the speed of rotation of the toothed-wheel from zero to ω, on further increasing the speed of rotation to 2ω, the light would appear again due to the passing of reflected light through the next slot. So, for every odd value of ω, light would disappear (stopped by tooth) and for every even value of ω light would appear (allowed by slot).

6.3.2 Speed of light through vacuum and different media

Scientists like Foucault (1819–1868) and Michelson (1852–1931) introduced different transparent media like glass, water etc., in the path of light to find the speed of light in different media. Even evacuated glass tubes were also introduced in the path of light to find the speed of light in vacuum. It was found that light travels with lesser speed in any medium than its speed in vacuum. The speed of light in vacuum was determined as, $c = 3 \times 10^8$ m s^{-1}. We could notice that the speed of light in vacuum and speed of light in air are almost same.

6.3.3 Refractive index

Refractive index **of a transparent medium is defined as the ratio of speed of light in vacuum c to the speed of light in that medium υ.**

$$\left.\begin{array}{r}\text{refractive}\\ \text{index } n \text{ of a}\\ \text{medium}\end{array}\right\} = \frac{\text{speed of light in vacuum } (c)}{\text{speed of light in medium } (\upsilon)}$$

XII_Physics_Vol-2_UNIT-6_EM.indd 12 26-12-2022 11:37:34